Homework 3: Requirement Analysis

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1. Entity Object

Client	Represents an individual or organization that requests event planning services.
Client Record	Stores all historical and current information about a client, including previous events and potential discounts.
Event	Represents a business event (e.g., workshop, conference, summer school) with details such as type, start/end dates, and preferences.
Request	Captures a client's request for organizing an event, with lifecycle states.
Task	Represents a work item assigned to sub-teams in the production or services department
Staff Record	Represents staff members, including their profiles, skills, and assignment history.
Schedule	Stores scheduling information for employees, events, and sub-teams to avoid conflicts.
Budget	Contains details of estimated, negotiated, and adjusted budgets for events and sub-tasks, possibly including discounts.
Report	Represents summary outputs such as employee utilization, client statistics, or financial reports.

2. Boundary Objects

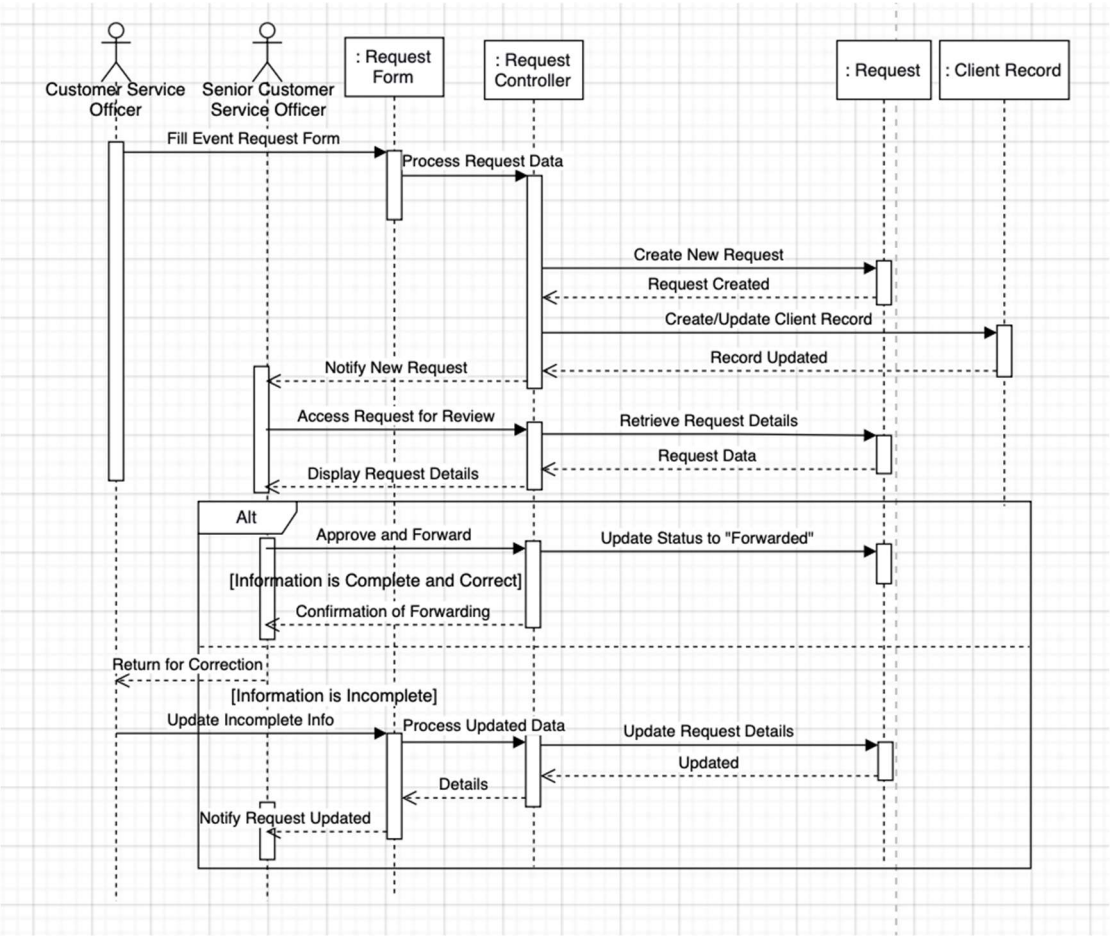
Request Form	Filled by customer service when recording the client's initial request for an event (event type, dates, preferences).
Recruitment Request Form	Used by production/service managers to request additional staff or outsourcing, submitted to the HR team.
Client Meeting Form	Filled during the first business meeting with the client by the senior customer service officer, recording client preferences and proposed budget.
Budget Request Form	Used by production/service managers to request budget changes, submitted to the financial manager.
Task List Interface	Interface where sub-team members (e.g., decorators, chefs, audio specialists) view

	and update their assigned tasks.
Report Interface	Interface used by administration, marketing, and vice president to generate and view reports

3. Control Objects

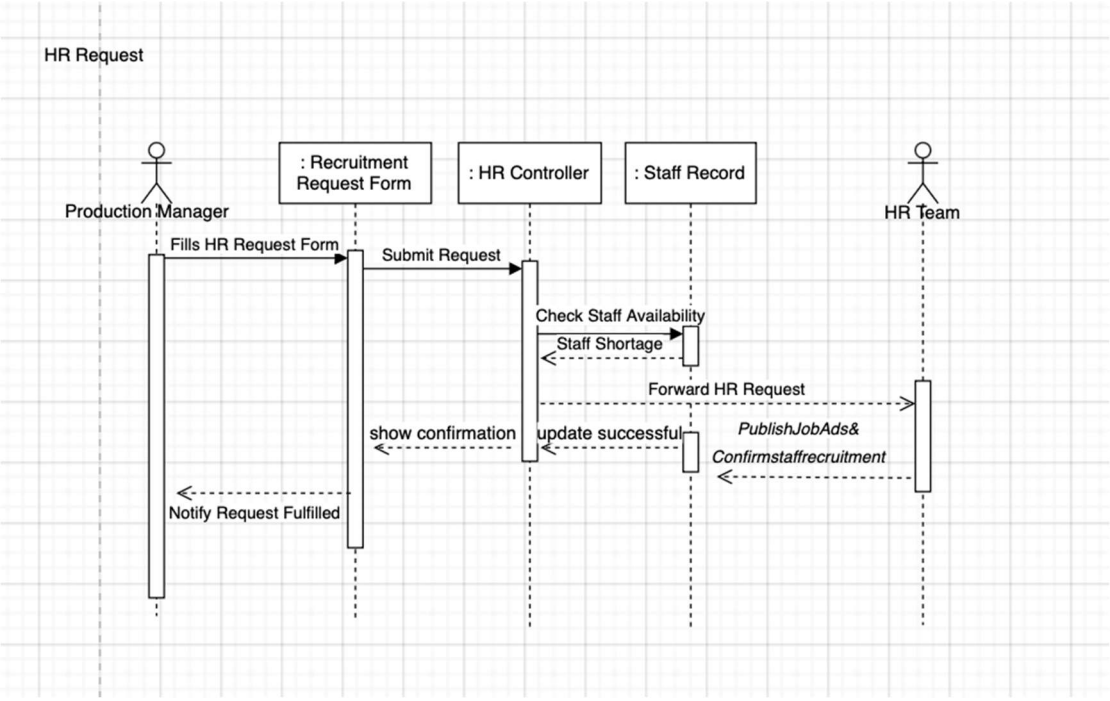
Request Controller	Handles the workflow of a new client request: receives data from the Request Form, validates it, updates the Request entity, and redirects it to the financial/administration managers for approval.
Meeting Controller	Manages the client meeting process: ensures preferences and proposed budget are recorded (via Client Meeting Form) and updates the Event and Budget entities accordingly.
HR Controller	Coordinates staffing requests: processes Resource Request Forms from production/service managers, forwards them to HR staff, and updates Staff Record entities when recruitment/outsourcing is confirmed.
Budget Controller	Handles budget adjustments , receives Budget Adjustment Forms, interacts with the Budget entity, and negotiates with the Financial Manager.
Task Assignment Controller	Manages task creation and distribution: production/service managers use it to assign tasks to sub-teams; updates Task entities and ensures they appear in the Task List Interface.
Report Controller	Orchestrates generation of summary and detailed reports (employees' utilization, client statistics, events earnings), linking Report Interface with underlying entities (Event, Staff Record, Client Record).

4. (1) Enter event request



This diagram shows how an event request is created and reviewed.
The Customer Service Officer submits the request form, the Request Controller creates the request and updates the client record, and the Senior Officer reviews it. If complete, the request is approved and forwarded; if incomplete, it is returned for correction and updated.

（2）HR request



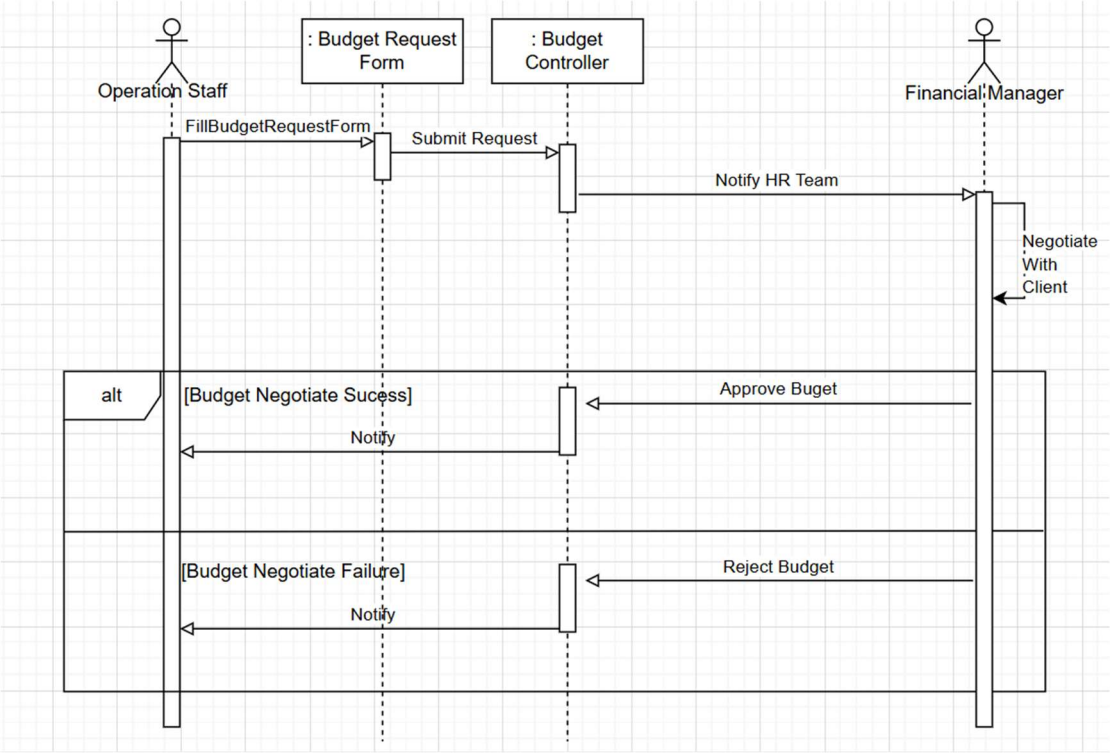
This diagram shows the process of handling an HR request.

The Production Manager submits a recruitment request form, which is processed by the HR Controller.

If staff shortage is found, the request is forwarded to the HR Team, who publish job ads and confirm recruitment.

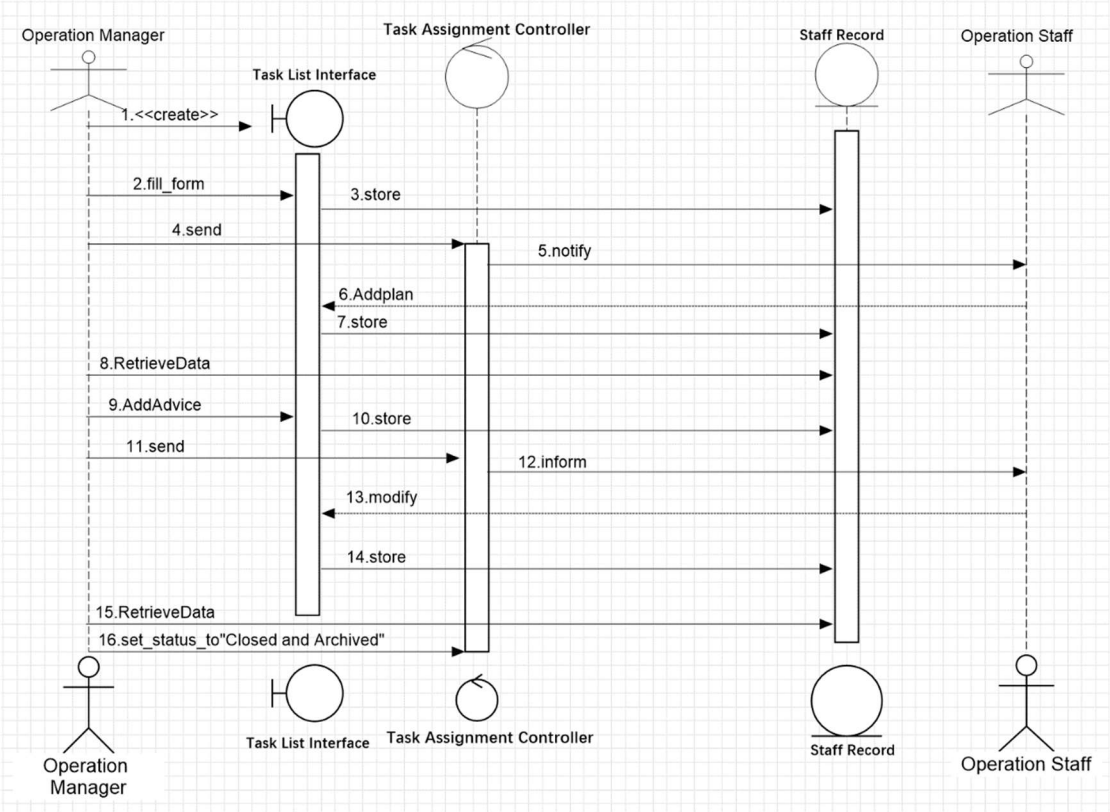
Once the records are updated, the Production Manager is notified that the request is fulfilled.

(3) Budget Adjustment



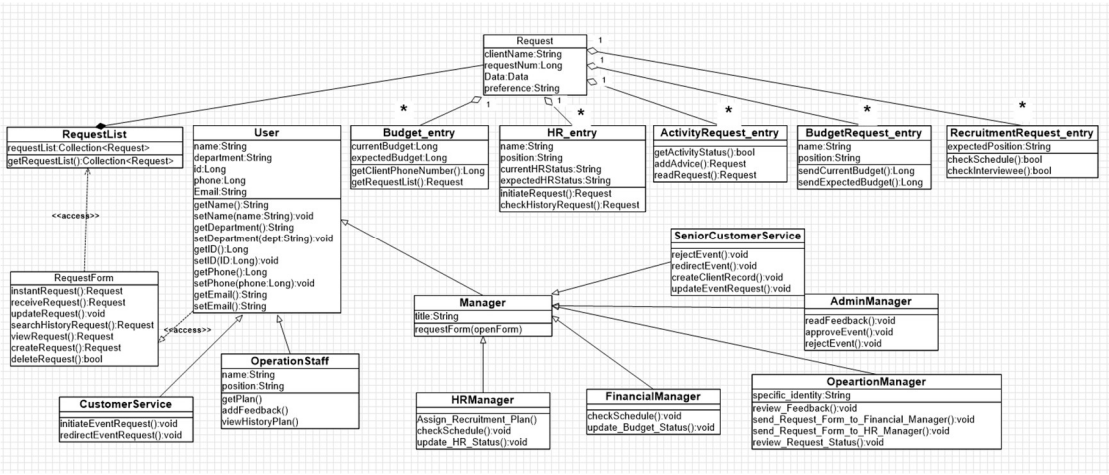
The budget adjustment request is initiated by the operation staff. The request will first create a form and activate the budget controller, which then notifies Financial Manager to handle the request. The sequence branches based on the result of negotiation between client and financial manager, and then financial manager approve/reject the request accordingly. The message will be send to operation staff by budget controller.

(4) Task Allocation



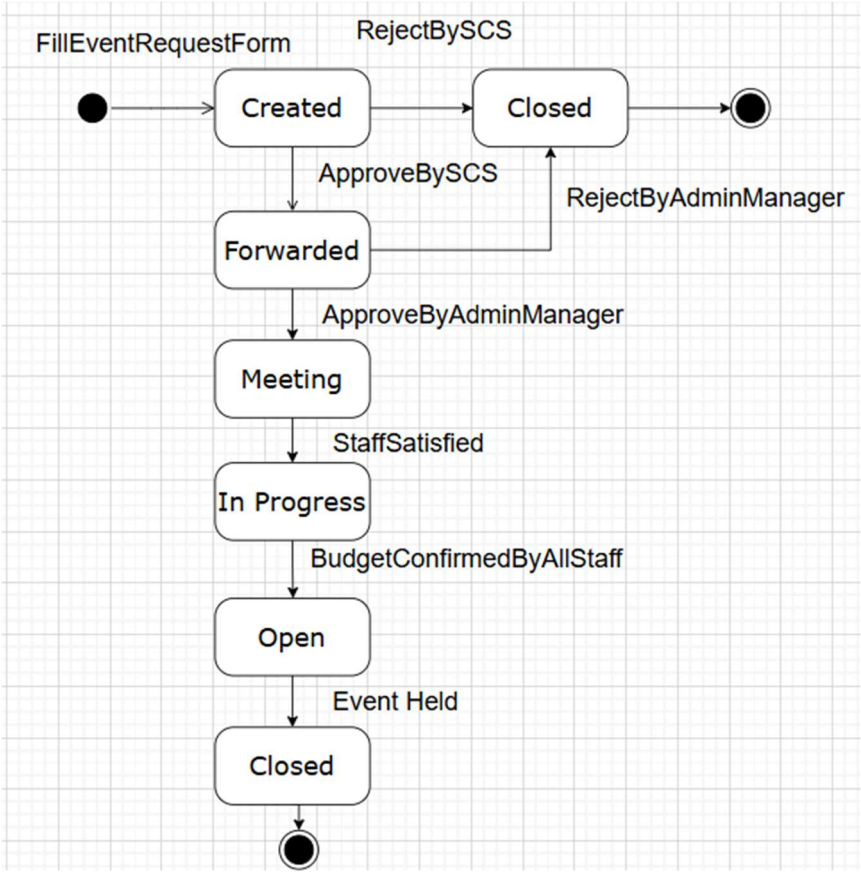
This diagram shows the process of task planning and allocation.
An Operation Manager initiates the process by creating and submitting a task through the Task List Interface. The Task Assignment Controller receives the task, stores it in the Staff Record, and notifies the designated Operation Staff. Throughout the process, the manager can add advice, and the controller can add plans or make modifications, with all updates being stored and communicated to the staff. To conclude the workflow, the Operation Manager retrieves the final task data and sets its status to "Closed and Archived."

5. Class Diagram

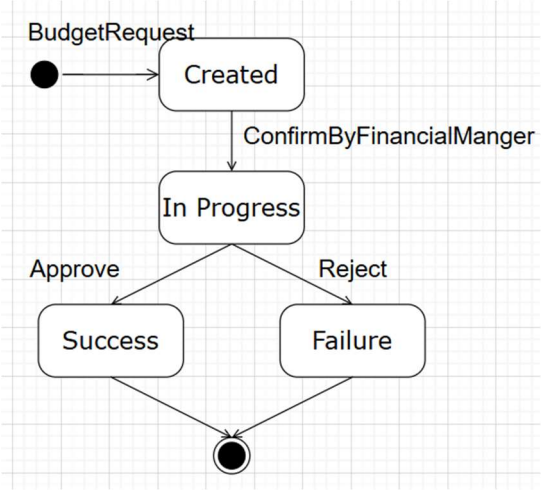


6. State Chart Diagram

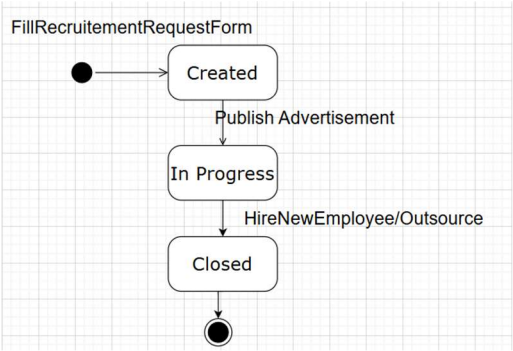
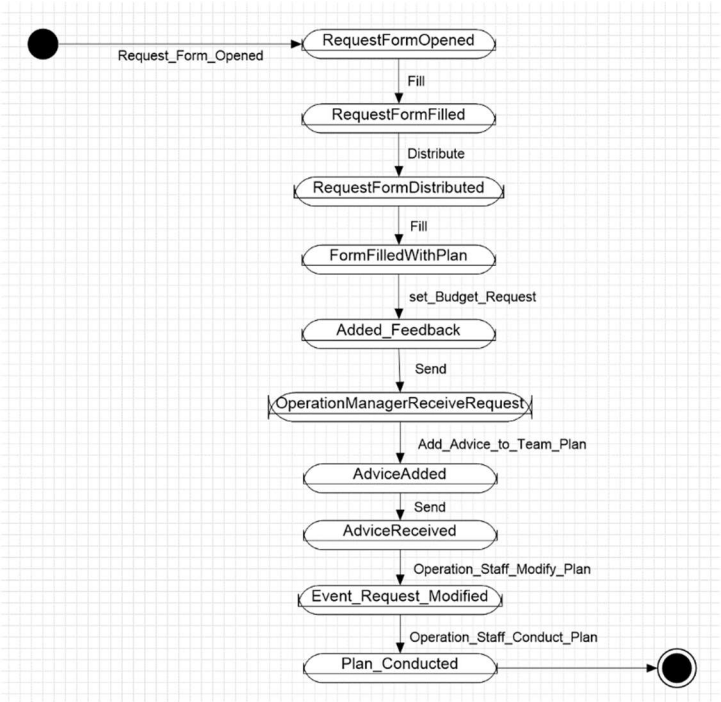
(1) Event Request (SCS = Senior Customer Service).



(2) Budget Request



(3) HR Request



(4) Task Status